

Project ID :

25-26J-311

1. Topic (12 words max)

Smart Seaweed Cultivation Assistant: IoT-Driven Environmental Monitoring and Farming Guidance System

2. Research group the project belongs to

TIM - Technology Integration and Management

3. Specialization of the project belongs to

Information Technology (IT)

4. If a continuation of a previous project:

Project ID	
Year	

5. Brief description of the research problem including references (200 – 500 words max) – references not included in word count.

Seaweed farming has gained global recognition as a sustainable form of aquaculture, contributing to food security, climate resilience, and income generation in coastal and island regions [1]. However, widespread adoption remains limited, particularly among small-scale and new farmers, due to environmental uncertainties, technical barriers, and lack of access to consistent cultivation support [2].

One major challenge is the absence of affordable and user-friendly tools for assessing environmental suitability. Seaweed is sensitive to fluctuations in water temperature, salinity, light intensity, and pH levels. Most farmers still rely on manual assessment or trial-and-error methods, which often result in inconsistent growth, reduced yield, or even crop failure [3]. For inland or home-based cultivators who depend on controlled environments, the inability to track these factors in real time makes seaweed farming more difficult and less reliable.

A second critical issue is the lack of structured guidance throughout the seaweed cultivation cycle. Farmers often do not receive step-by-step support for essential activities such as cleaning, sorting, monitoring growth, or determining harvest readiness. Without this guidance, harvests may be mistimed, reducing product quality and market potential. In addition, many existing aquaculture advisory tools are either not localized, not digitized, or too costly and complex for small-scale users to adopt [4].

These issues highlight a significant gap in the support available to seaweed farmers, especially those interested in using technology for improved monitoring and care. Bridging this gap could enable more informed, consistent, and scalable seaweed farming practices in both traditional and home-based environments.

- [1] Buschmann, A. H. et al. (2017). Seaweed production: overview of the global state of exploitation, farming, and emerging research activity. *European Journal of Phycology*, 52(4), 391–406.
- [2] Trivedi, N. et al. (2015). Seaweed farming: potential and challenges. *Aquaculture Reports*.
- [3] Springer. (2021). IoT-Based Aquaculture Monitoring Systems. In: *Advances in Intelligent Systems and Computing*.
- [4] Vairappan, C. S. (2017). Seaweed cultivation for sustainable development. *Marine Biotechnology*.

6. Brief description of the nature of the solution including a conceptual diagram (250 words max)

This research proposes a smart, IoT-integrated web application to improve the efficiency and accessibility of seaweed farming, particularly for small-scale and home-based cultivators. Traditional seaweed farming often relies on manual observation, which can result in poor decisions, low yields, or crop loss due to undetected environmental changes. The proposed system addresses these challenges by combining real-time sensor data with digital farming guidance.

IoT sensors are used to monitor critical environmental parameters such as temperature, salinity, pH level, and light intensity. These inputs help determine whether current conditions are suitable for seaweed growth. This is especially helpful for inland users cultivating seaweed indoors, as it allows them to simulate optimal marine conditions.

The system features a user-friendly web interface that provides real-time recommendations and visual insights based on sensor data. Farmers receive step-by-step care guidance and timely feedback, reducing reliance on manual expertise and improving cultivation outcomes. The platform also supports planning and preparation for each stage of the farming cycle.

By combining modern IT tools with aquaculture knowledge, the solution delivers accurate, data-driven support to seaweed farmers. It reduces guesswork, improves decision-making, and encourages broader adoption of seaweed farming, even in non-coastal regions.

This approach contributes to food security, environmental sustainability, and the growth of smart agriculture in under-resourced communities. (Conceptual diagram – Figure 1.0)

Smart Seaweed Farming IoT System

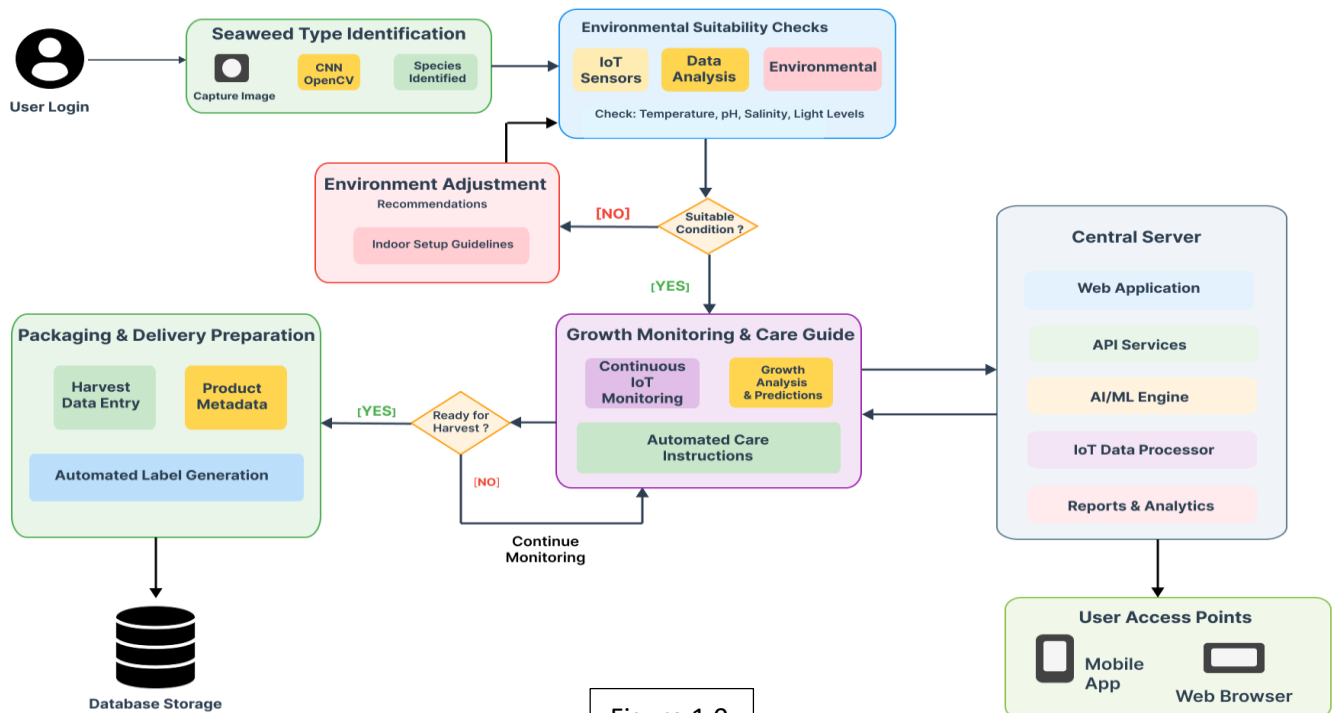


Figure 1.0

7. Brief description of specialized domain expertise, knowledge, and data requirements (300 words max)

Developing the proposed smart seaweed farming assistant requires expertise in IoT, web development, data processing, and seaweed cultivation. Knowledge of IoT is essential for collecting real-time environmental data using sensors for temperature, salinity, pH, and light. These sensors must be calibrated and connected to microcontrollers like Arduino or ESP32 to ensure reliable data transmission.

Full-stack web development skills are needed to build the system. The frontend will use tools like React.js, while the backend will be developed with Node.js and Express. A NoSQL database such as MongoDB will handle sensor data, user inputs, and system feedback. The interface should be user-friendly and accessible to farmers with varying technical skills.

Data processing skills are required to create logic that analyzes sensor inputs and triggers alerts when environmental conditions are unsuitable. Developers will also implement basic data visualization features to display trends and growth progress.

Domain knowledge in seaweed farming is equally important for defining accurate care instructions. Understanding species requirements, environmental conditions, cleaning methods, and harvest timing ensures the system offers meaningful guidance.

Key data requirements include seaweed species datasets, real-time sensor readings, and user-generated growth logs. Ensuring the accuracy, security, and usability of this data is vital for system performance and user trust.

8. Objectives and Novelty

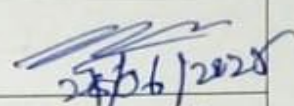
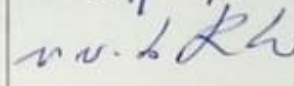
Main Objective To develop a smart, IoT-integrated, and user-friendly seaweed farming assistant system that supports species identification, environmental condition evaluation for indoor cultivation, growth monitoring, and packaging preparation to enhance quality and accessibility in sustainable seaweed farming.			
Member Name with Registration No	Sub Objective	Tasks	Novelty
IT22586452 Kulasinghe I.D	Seaweed Type Identification and Classification	• Create a digital database of seaweed species with images and key characteristics • Enable image upload or trait-based search by users • Display matching species with edibility, use, and warnings	Provides a simple identification interface for non-expert farmers using an integrated searchable knowledge base
IT22579454 Marasinghe M.A.S.S	Environmental Suitability Analysis for Indoor/Outdoor Growth	• Integrate IoT sensors to collect environmental data (light, pH, salinity, temperature) • Compare sensor data with species-specific ideal ranges • Suggest whether indoor or outdoor farming is viable • Recommend suitable seaweed types and container setups	First use of sensor-driven, real-time indoor/outdoor seaweed farming recommendations designed for small-scale or home growers

IT22250292 Peiris G.S	Cleaning, Sorting, Growth Monitoring, and Harvest Scheduling	• Guide cleaning and sorting steps based on seaweed quality • Allow users to log growth data (length, date) • Visualize growth with progress charts • Use IoT inputs to suggest optimal harvesting time	Combines manual tracking with sensor-based environmental data to predict harvest readiness and optimize seaweed quality
IT22113368 Dissanayake V.D.G	Packaging and Delivery Preparation	• Let users select packaging type (dried, fresh, bottled) • Input product metadata (label, weight, harvest date) • Generate printable labels and delivery preparation checklist	Provides a structured, guided packaging and traceability system for small-scale producers entering retail/export markets

9. Individual component description of how it is complied with the specialization.

Member Name with Registration No	Description
IT22586452 Kulasinghe I.D	This function involves developing a seaweed recognition feature that uses image data and characteristic matching to identify species. It integrates tools such as OpenCV for image processing and may use Convolutional Neural Networks (CNN) to classify seaweed types based on photos. The system stores and retrieves seaweed species data from a structured database, ensuring the application of database design, full-stack web development, and AI-based image classification. This complies with IT specialization through the application of artificial intelligence, computer vision, and backend systems.
IT22579454 Marasinghe M.A.S.S	This component uses IoT sensors (e.g., temperature, pH, salinity, light sensors) to collect environmental data. The system processes sensor readings using backend logic and suggests suitable growing conditions. The frontend will visualize data trends and feedback. Tools and technologies include Arduino/ESP32, Node.js, MongoDB, and data visualization libraries like Chart.js or D3.js. This function applies core IT skills in IoT integration, data processing, sensor networks, and real-time web applications.
IT22250292 Peiris G.S	This module guides users in maintaining seaweed cleanliness, tracking growth, and determining harvest timing. It relies on rule-based decision systems using growth logs and sensor inputs. Real-time data from IoT devices is visualized to detect anomalies or growth readiness. IT concepts applied include real-time data handling, automation, and user-guided analytics, supported by technologies such as Node.js, React, IoT modules, and simple ML logic for condition checking.
IT22113368 Dissanayake V.D.G	This feature provides a streamlined system for packaging and distributing farmed seaweed products, focusing on traceability and suitability for local consumption in Sri Lanka. It allows users to enter key product details such as harvest date, weight, seaweed type and processing method, and then generates printable labels, packing checklists and delivery summaries. The system also suggests edible seaweed product types. With support for CRUD operations and a user-friendly web interface built using React, Express.js and MongoDB, this feature ensures efficient post-harvest handling, documentation and supply management.

10. Supervisor details

	Title	First Name	Last Name	Signature
Supervisor	Mr	Hushanthy	SRUB	 24/6/2025
Co-Supervisor	Prof	Samath	Rajale	
External Supervisor				
Summary of external supervisor's (if any) experience and expertise				

This part is to be filled by the Topic Screening Staff members.

- a) Does the chosen research topic possess a comprehensive scope suitable for a final-year project?

Yes ☒ No ☐

- b) Does the proposed topic exhibit novelty?

Yes ☒ No ☐

- c) Do you believe they have the capability to successfully execute the proposed project?

Yes ☒ No ☐

- d) Do the proposed sub-objectives reflect the students' areas of specialization?

Yes ☒ No ☐

- e) Supervisor's Evaluation and Recommendation for the Research topic:

Seaweed is required project for
Vegem people in the country as well as
Export market

Acceptable: Mark/Select as necessary

Topic Assessment Accepted	
Topic Assessment Accepted with minor changes*	
Topic Assessment to be Resubmitted with major changes*	
Topic Assessment Rejected. Topic must be changed	

* Detailed comments given below

Comments

Staff Member's Name	Signature